Intelligent Disease Surveillance System

A Plain-English Guide

An Adaptive SVM Model for Intelligent Disease Surveillance

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1. In one sentence

It is a computer system that reads real disease-case records as they arrive, learns what an outbreak looks like, predicts where Lassa fever is likely flaring up, and automatically alerts the health officials in charge — and it **keeps getting smarter with every new case**, without ever being shut down to be retrained.

2. The problem it solves

- **Reporting is slow.** Outbreaks are often spotted only after they have already spread.
- **Models go stale.** A prediction model trained once on old data slowly drifts out of date as reality changes.
- **People notice too late.** By the time a human manually spots a spike in the numbers, people are already sick.

This system watches the data **continuously**, flags rising risk **early**, and tells the right people **immediately**.

3. The five jobs it does

Job	What it means
Diagnose	Given a suspected patient's symptoms, exposure and location, estimate whether it is really Lassa fever.
Predict outbreaks	For each state, forecast whether the coming period looks like an outbreak.
Assess severity	For a confirmed case, estimate whether it is high-risk (elevated death risk).
Send alerts	Automatically notify the people in charge the moment risk is high.
Show a dashboard	A live screen where officials see risk by state, triage cases, view trends, and read the alert log.

4. The “brain” — the Adaptive SVM

SVM = Support Vector Machine. A classic, well-trusted machine-learning method that draws the best possible dividing line between two groups — for example, “outbreak” versus “not an outbreak.”

“Adaptive” (online learning) is the special part. Instead of being trained once on a big pile of old data and then frozen, this model learns **one small batch at a time** and updates itself with every new case. Think of a nurse who gets sharper with each patient she sees — not one who studied years ago and then stopped learning.

Under the hood it is a **SGDClassifier(loss='hinge')** updated with **partial_fit()**. The “hinge loss” is what makes it an SVM; the partial_fit step is what lets it keep learning on the fly. **This always-learning ability is the core contribution of the thesis.**

5. The real data (not mock data)

- **Source:** the real SORMAS / NCDC Lassa fever line-list, published on Zenodo (record 7309567).
- **Size:** 20,062 real cases from 2018–2021, across 774 local government areas in 37 states.
- **Confirmed:** 3,162 of those cases were laboratory-confirmed positive.
- **Detail per case:** about 40 symptom flags, plus demographics, exposure history, outcome (recovered / deceased), dates and location.

These are actual Nigerian disease-surveillance records — not made-up numbers.

6. The results (told honestly)

Task	Accuracy (AUC)	What it means
Outbreak prediction	**~0.89 (strong)**	Correctly ranks Edo, Ondo and Bauchi — the real Lassa hotspots — as the highest-risk states.
Diagnosis from symptoms	~0.64 (modest)	Early Lassa symptoms look like malaria or flu, so symptoms alone are genuinely limited. This is a real, documented finding — not a bug.
Outcome / death risk	~0.57 (weak)	Reported honestly: symptoms are only a faint signal of who will survive.

The headline finding: the adaptive, always-learning model (0.89) is essentially as accurate as a traditional “train-once” batch model (0.91). In other words, you get the continuous-learning benefit at **almost no cost in accuracy** — which is exactly the point the thesis sets out to prove.

7. The tools, in plain English

These are deliberately lightweight stand-ins for the heavy “big-data” tools in the original design, chosen so the whole system runs on a single laptop.

Tool	Its plain-English role
Redis Streams	The conveyor belt — carries new case records into the system one after another.

Tool	Its plain-English role
Pandas (streaming loop)	The workbench — cleans and shapes each incoming batch of data.
MongoDB + SQLite	The filing cabinets — MongoDB holds the case records; SQLite holds the alert log.
Adaptive SVM (scikit-learn)	The brain — learns from each batch and makes the predictions.
Streamlit	The control-room screen — the dashboard officials actually look at.
Docker Compose	The moving box — packs the whole system so it runs anywhere with one command.

Why these? They replace the enterprise tools from the original plan — Redis for Kafka, Pandas for Spark, MongoDB + SQLite for Cassandra — so the entire system builds and runs on one laptop instead of a data centre.

8. What has actually been built

Main system — the Adaptive SVM (the thesis)

- Real SORMAS Lassa data loaded, cleaned and prepared.
- Three trained online SVMs — one each for diagnosis, outbreak prediction and outcome.
- A notification layer that logs every alert to SQLite and can e-mail the officials in charge.
- A 5-tab Streamlit dashboard: Overview, Outbreak monitor, Case triage, Trends, and Notifications.

Bonus system — the earlier build (still live online)

- An LSTM forecasting + anomaly-detection system covering four diseases (Lassa, cholera, meningitis, mpox).
- Fully deployed on the internet: React dashboard on Vercel, FastAPI backend on Render, database on MongoDB Atlas.
- **Live:** disease-surveillance-early-outbreak.vercel.app

9. The 30-second version (to say out loud)

> *"I built an intelligent disease-surveillance system for Nigeria. It uses an adaptive SVM — a machine-learning model that keeps learning from every new case instead of being trained once and going stale. Working on 20,000 real Lassa fever records, it predicts which states are heading into an outbreak with about 89% accuracy, and automatically alerts the health officials in charge. The novel part is that it learns continuously while running — and I proved it is just as accurate as a traditional model that has to be shut down and retrained."*

Prepared for the thesis: “An Adaptive SVM Model for Intelligent Disease Surveillance System.”